

QUOC ANH TRINH (ANDREW)

Data Analyst, M.Sc in Business Analytics, Deakin University



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DATA ANALYST

- Data Analyst with 5+ years’ experience in FinTech and banking, specialising in SQL- and Python-driven statistical analysis, transaction monitoring, and Power BI business intelligence dashboarding for leading institutions including Bendigo & Adelaide Bank and Remitano. Applies advanced statistical modelling, machine learning, NLP, and Generative AI / LLM solutions on AWS and GCP—building ETL pipelines and data-driven tools that reduced false positives by up to 67%, cut regulatory reporting time by 80%, and lifted marketing conversion by 30%. Passionate about translating complex data into measurable business outcomes across compliance, cost, and performance.

KEY SKILLS

AI Engineering: (Next.js, FastAPI, Python, SQL, Vector DB)	• Architected production-grade Agentic AI solutions (FastAPI/GCP) including typology extractors for framework comparison and enforcement action checkers for AML/CTF Program review
Advanced Analytics: (SQL, Python, R)	• Fine-tuned scenarios utilising optimisation algorithms & statistical calibration achieving 60-70% true positive rates, 67% alert reduction • Researched & designed 9 new detection scenarios, statistically calibrated to a ~70% false positive baseline for optimal detection breadth • Engineered SQL-based interim monitoring maintaining 100% continuity across 2M+ transactions • Designed recursive SQL for 150K+ MLM relationships reducing payout calculation complexity, eliminating graph database dependency • Replicated 40+ transaction monitoring scenarios in SQL for assurance & 100% logic alignment during Adelaide Bank migration
Visualisation: (PowerBI, Metabase)	• Engineered Power BI dashboards eliminating 70% frequent ad-hoc requests, reducing prioritisation time by 60% • Delivered 100% on-time analytical reports/dashboards with 8/10+ satisfaction
Regulatory Reporting: (SQL, LLMs, Python, NLP)	• Automated AUSTRAC reporting pipeline with SQL, Python, NLP and LLMs, reducing report preparation time by 80% and enhancing reporting quality.
Predictive Analytics & ML: (Python, PySpark, Scikit-learn)	• Automated email marketing via customer classification driving 30% CTR uplift
Data Engineering & ETL: (dbt, GCP BigQuery, Redshift, AWS Glue)	• Designed an AML workbench on GCP using dbt and BigQuery for external transaction monitoring model validation and scenario benchmarking • ETL optimisation reducing developer workload by 25% , cloud costs 30% , and dashboard runtime 70% • Built vendor-agnostic scenarios + automated workflows processing 150+ alerts daily (99.9% integrity)

PROFESSIONAL EXPERIENCE

Analyst	Bendigo & Adelaide Bank	05/2021 - Present
<i>Intelligence and Analytics - Financial Crime Risk</i> <i>Australian banking group with 4 commercial banks and 1 digital bank under management</i>		
• Own the transaction monitoring analytics function, serving operations, transformation, compliance, assurance, and governance teams.		
• Own the Power BI dashboards as part of the transaction monitoring product—designing, developing, and main-		

taining them to provide key measures, KPIs, and guardrails.

- Research and design new transaction monitoring detection scenarios, then build and fine-tune them using statistical and machine learning techniques.
- Perform statistical analysis to support scenario design, calibration, and fine-tuning.
- Research and deliver machine learning and NLP solutions for exploratory analysis and predicting SMR-triggered alert sets.

ACHIEVEMENTS

- **Decision Engineering:** Built vector search infrastructure and real-time streaming UI to power an AUSTRAC Knowledge Hub and **AML/CTF Program Review Hub**; integrated enforcement action statements to automatically detect and evaluate regulatory gaps.
- Led the establishment of an interim fraud and AML transaction monitoring function during Bendigo Bank's exit from DELPHI and transition from TechOne, engineering advanced SQL-based detection queries to maintain **100% monitoring continuity** across **2M+ transactions** with **zero compliance breaches**.
- **AI Enablement:** Co-designed and delivered an end-to-end **AI-driven UAR-to-SMR pipeline** implementing secure **data masking/rehydration** and **Pydantic-validated LLM prompting**; reduced regulatory reporting preparation time by **80%** while ensuring strict transparency and data privacy compliance.
- Developed and fine-tuned new transaction monitoring scenarios for Child Exploitation, Sextortion, Drug Trafficking, and Terrorism Financing typologies, leveraging genetic algorithms, statistical calibration, and trend modelling to achieve **60–70% true positive rates** with a **67% reduction** in alert volume.
- **Scenario Design:** Researched and designed **9 new** detection scenarios, applying statistical analysis to calibrate each to an effective **~70% false positive** baseline—maximising detection coverage while avoiding overly narrow scope.
- **Agentic AI Framework:** Engineered a **typology extractor** using Agentic AI and advanced ETL to transform **156 AUSTRAC guidance documents** into structured data for transaction monitoring **framework comparison** and national monitoring.
- Built complex SQL & Python-based transaction monitoring scenarios outside vendor platforms and engineered automated daily workflows to feed alerts alongside internal SMRs into vendor systems—processed **150+ alerts daily** with **99.9% data integrity**, eliminating manual transfers.
- Led the annual transaction monitoring performance review, applying statistical confidence testing, data sampling, and SQL-based scenario recalibration to reduce false positives by **5%** and remove **5,000+ unnecessary alerts annually**, saving **300+ investigation hours annually**.
- Engineered SQL-based replication and assurance testing of **40+ transaction monitoring scenarios** for the Adelaide Bank to Bendigo Bank migration, employing advanced validation queries and transformation logic to achieve **100% scenario alignment** and seamless model transfer.
- **AML Workbench Design:** Designed and developed an **AML analytics workbench** on **Google Cloud Platform** using **dbt** and **BigQuery** to support external transaction monitoring model validation; built modular, version-controlled data transformation layers for scenario testing, threshold calibration, and model performance benchmarking across the bank's financial crime detection pipeline.

Data Analyst

Remitano

03/2020 - 04/2021

International cryptocurrency exchange that matched \$500M+ in trading value and operated in more than 30 countries.

- Handled all data analytics requests, delivering dashboards and diagnostic reports using Python, SQL, and Metabase.
- Built marketing predictive models to automate and optimise strategy deployment.
- Designed, developed, and managed data crawlers, ETL pipelines, and Redshift database structures.
- Collaborated with engineering teams on complex SQL-based data transformations for scalable operational and marketing analytics.

ACHIEVEMENTS

- Automated email marketing via customer classification models (Python/SQL), boosting click-through rates by **30%** and enabling scalable campaign personalisation.
- Optimised ETL pipelines, indexing, and data normalisation in Redshift, slashing developer workload by **25%**, cloud costs by **30%**, and dashboard runtime by **70%**.

- Delivered **100% on-time** analytical reports and dashboards with consistent stakeholder satisfaction scores of **8/10 or higher**.
- Engineered complex SQL solutions (including recursive queries) for MLM commission calculations, processing **150K+ referral relationships** across **4 levels**—reduced payout computation complexity, eliminating the need for a graph database and powering a **4-tier referral program** that drove **25%** of new user growth.

Investment Associate

National Citizen Bank

09/2014-02/2018

National Citizen Bank – A Commercial Bank with more than 20 years of operation

- Built financial models for property project estimation, company valuation, and debt restructure planning, and presented these to the Investment Council for investment recommendations.
- Coordinated with in-house executives and clients in lending due diligence and gathering business information for financial analysis and valuation.

ACHIEVEMENTS

- Secured Investment Council approval for **80%** of recommendations, generating a public equity portfolio **CAGR of 10%**, drawing on field-tested financial analysis skills and public equity market experience.
- Reduced time for investment approval by **72 hours** and eliminated major conflicts in the investment process by drafting the investment procedure with detailed instructions for accounting journal entries, authority levels, and document revision.

HONOURS & AWARDS

- Third place & Honourable Mention - Project SafeAus - Making Australia safer, Team VAIT - GovHack Australia, 09/2021
- #13 in public test and #32 in private test - PetFinder Competition, Team Vicohub-1 – Kaggle, 02/2019-03/2019
- First prize & Runner-up - Programming Competition – Department of Education and Training – 2007, 2008 & 2009

PROFESSIONAL QUALIFICATIONS

- CFA Level 1 Passed – CFA Institute, 2013
- Google Cloud Platform Big Data and Machine Learning Fundamentals – Coursera, 2020
- Modernizing Data Lakes and Data Warehouses with GCP – Coursera, 2020
- Building Batch Data Pipelines on GCP – Coursera, 2020

EDUCATION

- Master of Business Analytics – Deakin University, 2019
- Bachelor of General Business Management – National Economics University, 2013

VOLUNTEER ACTIVITIES & HOBBIES

- Tutored 100+ students in SQL and Python
- Won 2 Open Gov hackathon prizes
- Served as Coordinator of CFA Vietnam volunteer team
- Hobbies: Sports (Football/Soccer, Powerlifting), Building data-driven solutions for micro-businesses

REFERENCES

Available upon request